

Program : Diploma in Communication and Computer Networking	
Course Code : 5321	Course Title: Software Management System
Semester: 5	Credits: 4
Course Category: Programme core course	
Periods per week: 4	Periods per semester: 60

Course Objectives:

- Understand Software Engineering Process and Process Models.
- Understand the importance of Project Planning
- Understand the Concept of Coding and Testing
- Understand the software project management activities

Course Prerequisites:

Nil

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the factors that drive software project and summarize the role of process model in a project	16	Understand
CO2	Summarize the concept of project Planning	13	Understand
CO3	Illustrate different coding principles with unit test process	16	Applying
CO4	To manage project schedule, risks, expenses, and resources with the application of suitable project management tools.	13	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	1	1	3	2
CO2	3	2	3	2	2	3	1
CO3	3	1	1	3	2	3	2
CO4	3	1	1	1	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the factors that drive software project and summarize the role of process model in a project		
M1.01	To understand Software Problems	4	Understanding
M1.02	To understand Software Process Models	6	Understanding
M1.03	To understand Software Requirement Specification	6	Understanding
Contents: Software Problem - Cost, Schedule and quality, Scale and Change. Software Process - Process and Project, Component Software Processes, Software development Process Models, Waterfall, Prototyping, Iterative development, Rational Unified Process, Spiral Model. Value of good SRS, Requirement process, Desirable Characteristics of an SRS, Components of an SRS.			
CO2	To Recognize the functioning of ARDUINO		
M2.01	To understand the concepts of Software Project Plan	7	Understand
M2.02	To understand the UML diagrams in Project Development	6	Understand
	Series Test –I	1	

Contents: Planning a software project - Effort Estimation, Project Schedule and Staffing, Quality Planning, Risk Management Planning, Risk Management Concepts, Risk Assessment, Risk Control, Project Monitoring Plan, Project Monitoring and tracking, Detailed Scheduling UML diagrams.			
CO3	To Understand different Coding principles with <u>Unit</u> test process		
M3.01	To understand the concept of Coding	8	Apply
M3.02	To understand the concept of Testing	8	Apply
Contents: Coding - Programming Principles and Guidelines, Coding Standards, An incremental coding process, Test Driven development, Unit Testing, Unit testing of Classes, Code Inspection, Planning, Group review meeting. Testing - Testing Concept, Error, Fault and Failure, Test Case, Levels of Testing, Testing Process, Test Plan			
CO4	To understand the concept of Software Configuration Management Project Management Tools		
M 4.01	To understand concept of Software Configuration Management.	10	
M 4.02	To understand concept of Project Management Tools	3	
	Series Test – II	1	
Contents: Software Configuration Management: Software Configuration Items and tasks, Baselines, Plan for Change, Change Control, Change Requests Management, Risk Management: Risks and risk types, Risk Management Process: Risk identification, Risk analysis, Risk planning, Risk monitoring, Software Project Management Tools: CASE Tools			

Text / Reference

T/R	Book Title/Author
T1	Software Engineering –A Precise Approach, Pankaj Jalote, edition 2010, Wiley India, ISBN: 9788126523115.
T2	Software Engineering A Practitioners Approach, Roger S Pressman, 2010, Tata McGraw Hill Publishing Co Ltd, ISBN 9780070701137
T3	“Project Management: The Managerial Process with MS” – Clifford F. Gray and Erik W. Larson, Mc Graw Hill
R1	Software Engineering 10 th edition, Ian Sommerville
R2	Software Engineering Principles and Practices, Rajesh Narang.Mc Graw Hill Education
R3	R. Mall, Fundamentals of Software Engineering
R4	Software Engineering, Sajan Mathew, 2009 Reprint, S Chand publications
R5	Software Project Management – Royce, Pearson Education

Online Resources

Sl.No	Website Link
1	https://www.educative.io/blog/software-process-model-types